

RACE Quiz Studio

A Reading Comprehension Quiz Generation System

CS Final Project | FAST NUCES | 2026

1. Abstract

This report describes the design and implementation of RACE Quiz Studio, an end-to-end reading comprehension quiz generator trained on the RACE dataset [1]. The system takes an English passage as input and produces a four-option multiple-choice question with graduated hints. It is built from traditional machine learning components, with a fine-tuned T5-small model included as a neural comparison baseline. The answer verification component (Model A) is a soft-vote ensemble of Logistic Regression, Calibrated SVM, and Random Forest classifiers operating on twelve hand-crafted TF-IDF and lexical features, reaching 36.5% MCQ accuracy against a 25% random baseline. The distractor generation component (Model B) is a Random Forest ranker trained on four features including edit distance and sentence-transformer cosine similarity, reaching 88% accuracy. A Streamlit interface ties both models into a step-by-step quiz experience with a developer dashboard that shows live metrics and compares traditional versus neural question generation outputs.

2. Introduction & Motivation

Generating meaningful reading comprehension questions automatically is a genuinely hard problem. The task requires identifying what facts in a passage are worth asking about, phrasing those facts as natural questions, and then producing plausible wrong answers that test real comprehension rather than keyword matching.

The RACE dataset [1] was constructed from real English examinations for Chinese middle and high school students. Unlike many reading comprehension datasets, the wrong options in RACE are written by human teachers to be plausible distractors, not random sentences. This means models trained on RACE have access to high-quality negative examples, which matters for both the answer verifier and the distractor ranker.

The motivation for this project was to understand what traditional ML alone can achieve on this task before reaching for large neural models. There is a tendency in NLP to jump straight to transformer fine-tuning, but the compute cost is high and results are not always interpretable. Building the feature engineering and ensemble from scratch gives a much clearer picture of which signals actually predict whether an option is correct.

3. Related Work

Du et al. [2] were among the first to frame question generation as a sequence-to-sequence task, training an attention-based encoder-decoder on SQuAD. Their model learned to identify answer-worthy spans and generate grammatical questions, though it required large amounts of labeled pairs. Zhao et al. [3] improved on this with paragraph-level generation using maxout pointers and gated self-attention, which helps the model stay consistent with broader context.

On the distractor side, Guo et al. [4] proposed generating distractors from real examination data using similarity and constraint-based retrieval. The intuition that good distractors come from the same domain as the correct answer carries through into our Model B design. Jiang and Lee [5] analyzed click-through data to measure distractor quality and found that edit distance and semantic similarity between distractor and correct answer are strong quality indicators.

BERT [6] and T5 [7] are the dominant pre-trained backbones for modern reading comprehension. We include T5-small fine-tuned on RACE as a neural baseline for question generation quality comparison. BLEU [8] and ROUGE [9] are standard automatic metrics for generated text, though both have known limitations when applied to open-ended question generation.

4. Dataset Analysis

4.1 RACE Overview

RACE [1] contains 97,687 questions across 27,933 English passages from Chinese middle and high school exams. Each CSV row has an article, a question, four options (A-D), and the correct answer label. We used the provided train.csv with 87,866 articles after deduplication.

4.2 Key Statistics

Metric	Value
Total articles (train)	87,866
Questions per article (avg)	~1.1
Avg article length (tokens)	~321
Avg question length (tokens)	~11
Answer label distribution	~25% each (A/B/C/D)
Avg option length (tokens)	~5.2
Training rows (after 1:1 balance)	149,372
Test rows (natural split)	52,720

4.3 Observations

- * Answer labels are roughly balanced across A, B, C, D with slight bias toward A in the middle school subset.
- * About 67% of questions are Wh-questions. The remainder are cloze-style fill-in-the-blank.
- * Correct answers appear verbatim in the article in approximately 58% of cases, making

exact-match a useful feature.

* Distractor options rarely contain the correct answer string, confirming exact-match is a strong negative signal.

* Article length follows a roughly log-normal distribution. High school articles exceed 600 tokens.

5. Model A: Answer Verification & Question Generation

5.1 Problem Framing

Model A has two jobs: (1) generate a question for an extracted answer candidate, and (2) score four answer options to identify the correct one. Job (1) is handled by a rule-based template system followed by an RF question ranker. Job (2) is binary classification: given a (article, question, option) triple, predict whether the option is the correct answer.

5.2 Answer Candidate Extraction

The system identifies answer candidates using regular expressions across six pattern types: dates, acronyms, possessive names, multi-word proper nouns, definition phrases (known as / called / referred to as), and location phrases (in/at/near [place]). Each candidate gets a heuristic score based on type and sentence position. The top eight are passed to the ranker.

5.3 Feature Engineering (12 features)

#	Feature	Description
0	tfidf_cosine_sim	TF-IDF cosine sim, article vs (question+option)
1	unigram_overlap_ratio	Fraction of (Q union O) tokens found in article
2	idf_weighted_score	Mean IDF weight of option tokens shared with article
3	best_sentence_overlap	Max unigram overlap over any single article sentence
4	bigram_overlap_ratio	Bigram version of feature 1
5	exact_match_flag	1 if option appears verbatim in article
6	option_token_length	Token count of option text
7	option_char_length	Character count of option text
8	first_position	Normalised first-occurrence position in article
9	is_number	1 if option is numeric or date-like
10	kmeans_cluster_label	K-Means (k=2) hard cluster assignment
11	kmeans_centroid_dist	Distance to nearest K-Means centroid

Feature 3 (best_sentence_overlap) is the single most discriminative feature. The correct answer typically aligns strongly with one specific sentence while distractors spread their overlap across many sentences. Features 10 and 11 come from an unsupervised step: TruncatedSVD(64) reduces option TF-IDF vectors, then K-Means(k=2) assigns cluster labels. These add a small but consistent improvement to ensemble F1.

5.4 Training Procedure

Training data: 87,866 RACE articles exploded into 4 option rows each, then 1:1 downsampled to 149,372 balanced rows. Test set: natural 75/25 split giving 52,720 rows (no resampling). Using `class_weight='balanced'` inside sklearn caused Calibrated SVM to predict all-negative. Manual 1:1 sampling before fitting produced better-calibrated probabilities for all models.

- * Logistic Regression: `C=1.0, max_iter=2000`
- * Calibrated SVM: `LinearSVC(C=1.0)` in `CalibratedClassifierCV(method='isotonic', cv=3)`
- * Naive Bayes: `GaussianNB` - trained and reported, excluded from ensemble vote
- * Random Forest: 300 trees, `max_depth=15, min_samples_leaf=5, max_features='sqrt'`
- * TF-IDF vectorizer: `max_features=20,000, sublinear_tf=True`

5.5 RF Question Ranker

A separate RF ranker is trained on 38 features per (question, answer candidate) pair, including question length, wh-word identity, template type (suffix_wh / definition / where_wh / cloze / generic), source sentence position, answer kind, and five Model A verifier scores. Trained on RACE gold (question, answer) pairs. Result: ROC-AUC 0.972, top-1 match rate 66.9% on held-out set.

5.6 Neural Baseline: T5-small

For comparison, T5-small was fine-tuned on 20,000 RACE Wh-questions for 5 epochs (batch 16, `lr=3e-4`). Input: 'generate question: context: <article> answer: <answer>'. The model generates fluent questions but occasionally produces generic outputs that could fit many articles. Template+ranker questions are less fluent but more specific on average.

5.7 Results

36.5%	57.9%	0.376	0.526 +/- 0.002	149,372
MCQ Accuracy	Binary Accuracy	Binary F1	Cross-Val F1	Training Rows

Classifier	In Vote	Accuracy	Precision	Recall	F1
Logistic Regression	Yes	0.5689	0.2907	0.5031	0.3685
Calibrated SVM	Yes	0.5746	0.2917	0.4913	0.3661
Naive Bayes	No	0.6579	0.3121	0.3058	0.3089
Random Forest	Yes	0.5864	0.3043	0.5086	0.3808
Ensemble (LR+SVM+RF)	--	0.5788	--	--	0.3762

Confusion matrix on 52,720 test rows: TP=6,698 FP=15,726 FN=6,482 TN=23,814. MCQ accuracy of 36.5% is 11.5 points above the 25% random baseline. Naive Bayes is excluded from the vote because its recall (~30%) is much lower than LR/RF (~50%), which suppresses the correct option's soft-vote score and hurts MCQ ranking.

6. Model B: Distractor Generation & Hint System

6.1 Overview

Model B supplies three plausible wrong answer options and three graduated hints. Candidates come from an offline-built index of 87,866 RACE articles. At inference time the retriever finds options from other questions sharing the same answer category (date, person, acronym, proper noun, etc.), filters by compatibility rules to avoid semantic leakage, then ranks them with a trained Random Forest.

6.2 Distractor Ranker Features (4 features)

Feature	Description
Edit distance	Levenshtein distance between correct answer and candidate
Length difference	Absolute character length difference
In-article flag	1 if candidate text appears in the source article
Cosine similarity	Sentence-transformer (all-MiniLM-L6-v2) cosine sim to correct answer

6.3 Training

Training data: 3,000 sampled RACE rows. For each row, the three non-correct options are labeled positive (good distractors, label=1) and a set of WordNet generic terms (object, thing, unknown, item) are labeled negative (bad distractors, label=0). Random Forest (150 trees, max_depth=12) was compared against a Logistic Regression baseline.

6.4 Results

88.0%	85.0%	82.0%	0.83
RF Accuracy	RF Precision	RF Recall	RF F1

6.5 Hint Generation

Three graduated hints are generated entirely by rule, with no ML at inference time. This was intentional: rule-based hints are fast, deterministic, and easy to debug.

- * Hint 1 (General): First sentence in the article that does not mention the answer.
- * Hint 2 (Intermediate): Sentence immediately before the answer sentence.
- * Hint 3 (Specific): Answer sentence with the answer replaced by '[BLANK]'.

7. User Interface

The front end is a Streamlit app (ui/app.py) with four screens managed through session state. A background daemon thread pre-loads the pipeline on first page visit so the UI is immediately responsive and the user never sees a cold-start delay. The pipeline singleton is cached via @st.cache_resource, so subsequent calls return instantly.

- * Screen 1 - Article Input: paste text, load a built-in sample, or pick from 500 RACE rows.
- * Screen 2 - Quiz View: generated question with four radio options. After answering, shows verifier probability scores for all four options as a progress bar table.

- * Screen 3 - Hint Panel: progressive hint reveal (Hint 1 -> 2 -> 3 -> Reveal Answer).
- * Screen 4 - Developer Dashboard: latency tile, confusion matrix, per-classifier table, T5 vs template question comparison, and session log CSV export.

8. Evaluation & Discussion

8.1 Answer Verification

The 36.5% MCQ accuracy is above random (25%) but well below human performance on RACE (~94%). This is expected: twelve lexical features cannot handle inference questions, which make up a large fraction of RACE. The ensemble mainly catches factual extraction questions where the correct answer appears verbatim or near-verbatim in the article.

The 5-fold cross-validation F1 of 0.526 is higher than the held-out test F1 of 0.376. The gap comes from distribution shift: cross-val runs on the balanced training data while the test set uses the natural 25/75 class ratio. MCQ accuracy better reflects real task performance because it asks whether the highest-scored option is correct, not whether the binary threshold is tuned correctly.

8.2 Question Generation Quality

No automated BLEU/ROUGE evaluation was run against gold RACE questions. Manual review of 50 generated questions showed `suffix_wh` templates produce grammatical questions ~72% of the time. Cloze questions are always grammatical but feel mechanical. T5 outputs are more fluent but occasionally hallucinate names not present in the passage.

8.3 Distractor Quality

The 88% distractor ranker accuracy measures whether a candidate beats a generic WordNet term, not whether it would fool a human. Qualitatively, distractors retrieved from the same category are usually plausible. Mismatches slip through when the article is short and the retrieval pool is small, causing the system to fall back to passage candidates from different categories.

9. Limitations & Future Work

- * MCQ accuracy is bottlenecked by features. Adding NER type matching, dependency parse features, or a sentence embedder would meaningfully improve ranking.
- * The distractor ranker trained on only 3,000 articles. Full dataset training would improve quality.
- * Inference questions requiring cross-sentence reasoning are essentially unsolvable with lexical features. A BERT entailment model would handle these cases.
- * No quantitative BLEU/ROUGE evaluation of question quality. This should be added.
- * Hint generation is rule-based and can reveal too much or too little depending on article structure.
- * The distractor index covers only the training split. A broader corpus would increase option diversity.

10. Conclusion

We built a complete reading comprehension quiz generation pipeline using traditional ML, with a neural baseline for comparison. The system extracts answer candidates with regex, generates template questions, ranks them with a trained RF ranker, retrieves and ranks distractors from the RACE corpus, and produces three graduated hints. The answer verifier ensemble reaches 36.5% MCQ accuracy (11.5 points above random) on the RACE test set. The distractor ranker reaches 88% accuracy distinguishing good distractors from generic terms.

The biggest lesson from this project is that feature engineering matters. The `best_sentence_overlap` feature alone accounts for a large share of the ensemble's discriminative power. Traditional ML with well-designed features remains a reasonable baseline for this kind of task, and the interpretability is a genuine advantage over neural approaches.

11. References

- [1] Lai, G., Xie, Q., Liu, H., Yang, Y., & Hovy, E. (2017). RACE: Large-scale ReAding Comprehension Dataset From Examinations. EMNLP 2017. <https://aclanthology.org/D17-1082>
- [2] Du, X., Shao, J., & Cardie, C. (2017). Learning to Ask: Neural Question Generation for Reading Comprehension. ACL 2017. <https://aclanthology.org/P17-1123>
- [3] Zhao, Y., Ni, X., Ding, Y., & Ke, Q. (2018). Paragraph-level Neural Question Generation with Maxout Pointer and Gated Self-attention Networks. EMNLP 2018. <https://aclanthology.org/D18-1424>
- [4] Guo, Q., Zhu, X., Chao, W., & Ye, Z. (2016). Generating Distractors for Reading Comprehension Questions from Real Examinations. AAAI 2016. <https://ojs.aaai.org/index.php/AAAI/article/view/10395>
- [5] Jiang, Y., & Lee, H.-Y. (2017). What Click-Through Data Can Tell Us About Distractor Quality. BEA Workshop 2017. <https://aclanthology.org/W17-5019>
- [6] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL 2019. <https://aclanthology.org/N19-1423>
- [7] Raffel, C. et al. (2020). Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. JMLR 21(140). <https://jmlr.org/papers/v21/20-1166.html>
- [8] Papineni, K. et al. (2002). BLEU: a Method for Automatic Evaluation of Machine Translation. ACL 2002. <https://aclanthology.org/P02-1040>
- [9] Lin, C.-Y. (2004). ROUGE: A Package for Automatic Evaluation of Summaries. ACL Workshop 2004. <https://aclanthology.org/W04-1013>